

SIDHARTH VIJAYAN

3rd-year Computer Engineering Student

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PROFESSIONAL SUMMARY

A results-driven pre-final year Computer Engineering student (MIT WPU) with a comprehensive foundation in Python, Machine Learning, and Data Analysis. Demonstrated capability in leading technical initiatives as the Synapse AI Club Management Head, combined with practical experience in automation and full-stack project development. Seeking an internship to translate complex data into actionable solutions.

WORK EXPERIENCES

Management Head and Technical Member

Aug 2024 - Present

Synapse AI Club, MIT WPU

- Instructed and mentored students in fundamental concepts through beginner bootcamps and specialized workshops.
- Coordinated organizational events, including technical workshops, guest lectures, and hackathons, driving member engagement and skill development.
- Drove digital strategy and scaled social media presence from a dozen to 700+ followers across LinkedIn and Instagram (~50x growth), significantly improving club visibility.

Technologies / Skills Used : Python, OpenCV, MediaPipe, Pandas, Numpy, Scikit-Learn, Pygame, Streamlit

PROJECTS

PolyRAG — Polymorphic Retrieval-Augmented Generation System | Github Website

- Architected a 5-agent RAG pipeline ingesting PDFs, Excel, and images into ChromaDB vector collections with top-4 semantic search via sentence-transformers embeddings.
- Engineered a 3-tier LLM fallback chain (Groq Llama 3.1 70B → Gemini 1.5 Flash → Ollama) with auto-failover, ensuring zero-downtime response generation under rate limits or outages.
- Extended with Context Police, a RAG auditing module evaluating retrieval relevance, hallucination risk, and context coverage — with a self-healing loop for query rewriting and answer patching.

Technologies / Tools Used : Python, LangChain, ChromaDB, Google Gemini API, Ollama, Streamlit, sentence-transformers, PyMuPDF

CareerCopilot — AI-Powered Resume Intelligence & ATS Optimization Platform | Github

- Built a full-stack AI app with 5+ LLM workflows — ATS scoring, skill-gap detection, bullet rewriting, role tailoring, project suggestions, and cover letter generation.
- Designed a production-grade Next.js + PostgreSQL architecture with Supabase auth, Prisma ORM, schema validation, protected routes, and modular APIs for scalable deployment.
- Delivered a SaaS-style dashboard with keyword heatmaps, PDF/DOCX parsing, drag-and-drop uploads, and reusable TypeScript components — with dark/light theming and smooth UX.

Technologies / Tools Used : Next.js 15, TypeScript, Tailwind CSS, Groq API, Prisma, PostgreSQL, Supabase

Crop Disease Detection and Future Risk Prediction System | Github

- Built an EfficientNet-B4 deep learning pipeline achieving ~96–98% accuracy across 38 disease classes on 54,305 PlantVillage images with a custom classifier head.
- Boosted real-world robustness via PlantDoc domain adaptation and Grad-CAM explainability; added confidence gating for reliable production-grade inference.
- Fused CNN image embeddings with an LSTM weather time-series model for joint disease detection and yield prediction; deployed via FastAPI, Streamlit, and Flutter with ONNX export.

Technologies / Tools Used : Python, PyTorch, EfficientNet, LSTM, FastAPI, Flutter, OpenCV, Grad-CAM

SKILLS

Databases :	MySQL, PostgreSQL, ChromaDB
Frameworks Libraries :	NumPy, Django, Flask, Matplotlib, Power BI, Pandas, React, FastAPI
Programming Languages :	Python, C++, Java, JavaScript, R, SQL
Tools Platforms :	Git, SQL, GitHub, Docker, LangChain, Streamlit, CI/CD, Postman

EDUCATION

MIT World Peace University

Aug 2023 - Present

Bachelor of Technology - Computer Science Engineering - AI & Data Science

CGPA : 8.18

GEMS OUR OWN ENGLISH HIGH SCHOOL | Sharjah, UAE

Apr 2022 - May 2023

Senior Secondary School

Grade : 82.8

Leaders Private School | Ajman, UAE

Apr 2020 - Mar 2021

Secondary School Certificate

CGPA : 89.8